

ROODY POINTJOUR

CE PROJECT PORTFOLIO

Embedded Software Engineer / Computer Engineer

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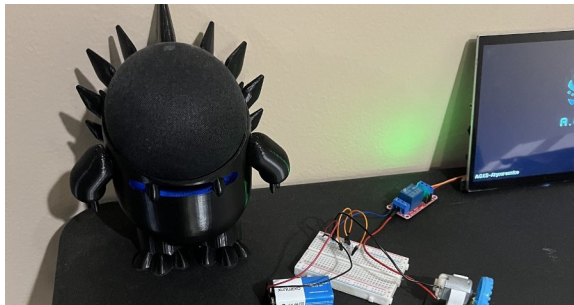
rpointjour.github.io/digital_business_card

Alexa Raspberry Pi IoT

IOT / VOICE CONTROL

Overview

The goal of this project was to build a custom Alexa Skill that can remotely trigger a relay-controlled DC motor connected to a Raspberry Pi. Voice requests are handled by the Alexa Skills Kit and routed to a Flask server running on the Pi, which is exposed to the internet through a Cloudflare Tunnel behind a custom domain, so the skill can reach the device securely without any port forwarding.



Hardware

- Raspberry Pi (Linux / Debian)
- Relay Module
- DC Motor

Software

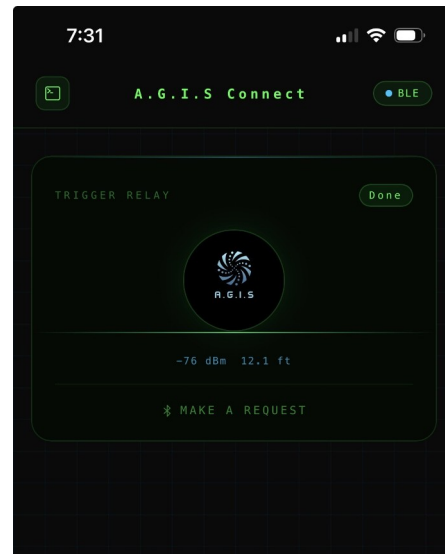
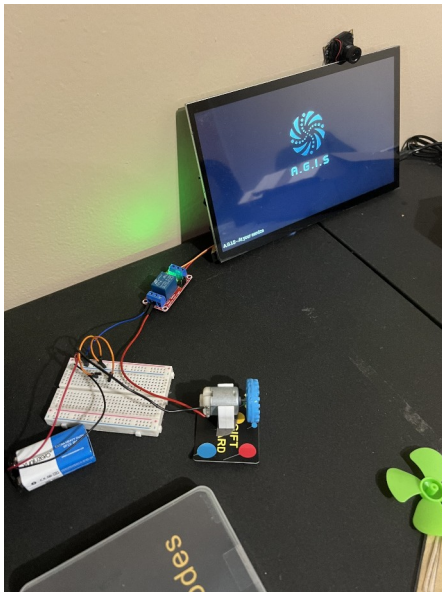
- Flask (Python)
- Alexa Skills Kit (ASK SDK)
- Cloudflare Tunnel
- Porkbun DNS

Raspberry Pi BLE Server & Client App

EMBEDDED SYSTEMS

Overview

The goal of this project was to build a Bluetooth Low Energy (BLE) system linking a Raspberry Pi to native phone and watch client apps. The Pi runs as a BLE GATT server, advertising services that a companion iOS, watchOS, or wearOS client discovers and connects to. The client sends requests over BLE GATT to trigger a relay-controlled DC motor, with live signal strength and distance feedback shown in the app.



Hardware

- Raspberry Pi (Linux / Debian)
- Relay Module
- DC Motor
- Breadboard + Touchscreen Display

Software

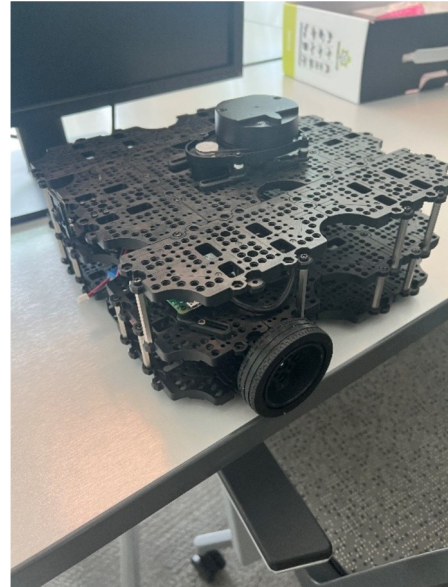
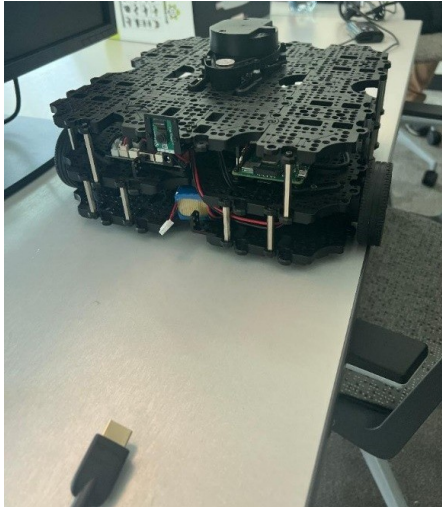
- Python (BLE GATT Server)
- Swift / Xcode (iOS, watchOS Client)
- Java / Android Studio (Android, wearOS Client)

TurtleBot3 (Waffle Pi)

ROBOTICS / ROS

Overview

The goal of this project was to design and build a robot capable of autonomously following the dynamics of a leader. I worked together alongside my team to program the robot to follow a human – the follower in the leader-follower system could track the trajectory of the leader without any collisions.



Hardware

- Remote Controller (Bluetooth, BLE)
- 360 Degree Laser Distance Sensor (for SLAM & Navigation)
- Sensors: Gyro, Accelerometer, IR Sensor, Motion Sensor, Distance Sensor

Software

- ROS - Robot Operating System
- SLAM - Used for mapping

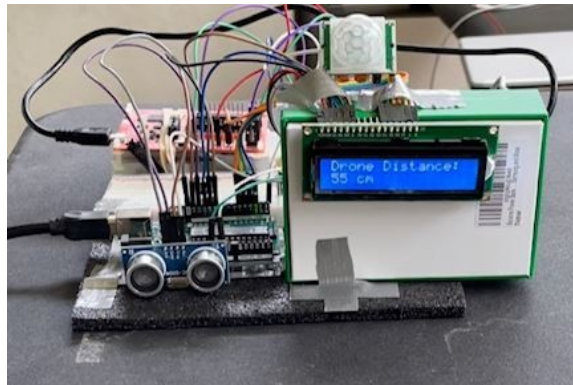
Distance Reader

EMBEDDED SYSTEMS

Overview

The goal of this project was to build a lightweight embedded system that measures real-time distance using an ultrasonic sensor and displays the reading live. It's a compact, ground-up exercise in low-level embedded programming – from sensor timing and signal processing to driving the output display – all running on Arduino.

► **Demo video:** <https://www.youtube.com/watch?v=w-XhrnkmF6g>



Hardware

- Arduino
- Ultrasonic Sensor

Software

- Embedded C